

Algebra II with Trigonometry

Chapter 5.5

Mental Math

Saved tutoring response with MathJax rendering. Original PDF file:
[4465374_alg-2trig-h-chp-5-5-note-docx-1.pdf](#)

Here are **10 mental math questions** based on the Law of Sines concepts in the PDF. All questions are designed to be answerable without paper or calculator, focusing on understanding rather than calculation.

1. Core Concept

Q: In any triangle, what are the sides proportional to? > **A:** The sines of their opposite angles.

2. Formula Recall

Q: State the Law of Sines in the form relating sides and sines of angles. > **A:**

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

3. Triangle Cases

Q: If you know two angles and one side of a triangle (ASA or SAA), which Law would you use to solve for unknowns? > **A:** The Law of Sines.

4. Ambiguous Case (SSA)

Q: In the ambiguous (SSA) case, what must you compare the given side to first to check if a triangle is possible? > **A:** Compare it to the height $h = b \sin A$.

5. No Solution Test

Q: In an SSA triangle where side $a < h$, how many triangles exist? > **A:** No solution (no triangle).

6. Angle Calculation

Q: The sum of the angles in any triangle is always what? > **A:** 180° .

7. Double Solution

Q: What is one sign that two triangles are possible in the SSA case? > **A:** Both the angle and its supplement less than 180° lead to valid triangles.

8. Right Triangle Check

Q: If one of the angles in a triangle is 90° , what type of triangle is it? > **A:** Right triangle (not oblique).

9. More Than One Triangle?

Q: If you know two sides and an angle not between them (SSA), could you always get exactly one triangle? > **A:** No, you could get 0, 1, or 2 triangles.

10. Formula Comparison

Q: Does the Law of Sines work for all triangles, or only for right triangles? > **A:** Works for all triangles (not just right triangles).

Let me know if you want harder or calculation-based mental math questions for practice!